

Tessa Frincke

Research Interests

Minor Bodies, Solar System Dynamics, Interstellar Objects, Exoplanets

Education

- 2025- **Ph.D.**, *Michigan State University, Department of Physics and Astronomy*, East Lansing, MI.
◦ Ph.D. Advisor: Professor Darryl Seligman
- 2021-2025 **B.S.**, *University of Toledo*, Toledo, OH., Physics, Concentration Astrophysics, Magna Cum Laude.
◦ Senior Honors Thesis: "A PAH Investigation of Star Clusters in the Antennae Galaxies NGC 4038/39"
◦ Thesis Advisor: Professor Rupali Chandar

Teaching Experience

- 2025- **Graduate Teaching Assistant**, *Michigan State University*, East Lansing, MI.
◦ Visions of the Universe Laboratory: Leads, plans, and manages lab section, gives short lecture at the beginning of lab, assists non-major students in lab, and grades weekly lab work.
◦ Studio Physics for Life Scientists I: Holds weekly office hours, assists pre-med students in class, and grades lab reports.

Honors And Awards

- 2025 Jesup Scott Honors College Medallion, University of Toledo, Toledo, OH
2024 NSF REU, University of Michigan, Ann Arbor, MI
2021-2025 Dean's List for 8 semesters, University of Toledo, Toledo, OH

Leadership and Service

- 2024-2025 Vice President, Society of Physics Students, University of Toledo, Toledo, OH

Selected Grants & Observing Proposals

- 2025 Michigan-Dartmouth-MIT Observatory PI Salazar Manzano, A11pl3Z, Chasing the third interstellar object 3I ATLAS, co-I, 10 nights

Observational Experience

- 09/2025 Remote Observations of 3I/ATLAS, Southern Astrophysical Research (SOAR) Telescope, Cerro Pachón, Chile, 3 half-nights, Lead Observer

- 08/2024 Patroclus-Menoetius Occultation Campaign, Southwest Research Institute, Boulder, CO, 4 nights
- 10/2024 Student Observing Campaign, Lowell Discovery Telescope, Flagstaff, AZ, 1 night

First Author Publications ([ADS Link](#))

2. **Frincke, T.** and Seligman, D., "Assessment of the Mass Loss and Radius Change of 3I/ATLAS Based on Observed Production Rates." *MNRAS*, Submitted, 2026. ([Link](#))
1. **Frincke, T.**, Yaginuma, A., et al. "Near-Discovery SOAR Photometry of the Third Interstellar Object: 3I/ATLAS." *MNRAS*, Accepted, 2025. ([Link](#))

Other Publications

4. Salazar Manzano, L., Lin, H., Tyler, A., Seligman, D., Adams, F., Gerdes, D., Ruch, T., **Frincke, T.**, and Napier, J. , "Onset of CN Emission in 3I/ATLAS: Evidence for Strong Carbon-Chain Depletion." *ApJL*, Accepted, 2025. ([Link](#))
3. Yaginuma, A., **Frincke, T.**, Seligman, D., Mandt, K., DellaGiustina, D., Peña-Asensio, E., Taylor, A., Nolan, M., "The Feasibility of a Spacecraft Flyby with the Third Interstellar Object 3I/ATLAS from Earth or Mars." *ApJ*, Accepted, 2025. ([Link](#))
2. Seligman, D., Micheli, M., Farnocchia, D. and 44 coauthors including **Frincke, T.**, "Discovery and Preliminary Characterization of a Third Interstellar Object: 3I/ATLAS." *ApJL*, Accepted, 2025. ([Link](#))
1. Salazar Manzano, L., Gerdes, G., and 6 coauthors including **Frincke, T.**, "Prospects of a New L5 Trojan Flyby Target for the Lucy Mission." *PSJ*, Accepted, 2025. ([Link](#))

Invited Conference Presentations

- 01/2026 **Frincke, T.**, "Near-Discovery SOAR Photometry of the Third Interstellar Object: 3I/ATLAS," Oral presentation, Invited Early Career Speaker at the 34th Meeting of the NASA Small Bodies Assessment Group (SBAG), Johns Hopkins University, Baltimore, MD.

Presentations

- 11/2025 **Frincke, T.**, "Near-Discovery SOAR Photometry of the Third Interstellar Object: 3I/ATLAS," Poster presentation at the Great Lakes Exoplanet Area Meeting (GLEAM), University of Wisconsin-Madison, Madison, WI.
- 02/2025 **Frincke, T.**, "Characterization of Star Clusters in the Antennae Galaxies NGC 4038/39," Poster presentation at the Jesup Scott Honors College Open House Dinner, University of Toledo, Toledo, OH.
- 01/2025 **Frincke, T.**, "Simulating and Identifying Encounter-able Hilda Asteroids," Poster presentation at Conference for Undergraduate Women and Gender Minorities in Physics (CU*iP), Michigan Technological University, Houghton, MI.

- 12/2024 **Frincke, T.**, "Simulating and Identifying Encounter-able Hilda Asteroids," Poster presentation at Undergraduate Research and Creative Activity Exhibition, University of Toledo, Toledo, OH.
- 11/2024 **Frincke, T.**, Stewart, C., and Beach, K., "Investigation of Star Clusters in the Arp220 Galaxy," Oral presentation at Ritter Symposium, University of Toledo, Toledo, OH.
- 10/2024 Gerdes, D., Levison, H., Napier, K., Salazar Manzano, L., Lin, J., and **Frincke, T.**, "Spacecraft encounters with resonant populations: an optimal target search strategy," Oral presentation at AAS Division for Planetary Sciences meeting 56, id. 209.04. Bulletin of the American Astronomical Society, Vol. 56, No. 8 e-id 2024n8i209p04
- 08/2024 **Frincke, T.**, "Simulating and Identifying Encounter-able Hilda Asteroids," Oral presentation at NSF REU Symposium, University of Michigan, Ann Arbor, MI.